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How to Install the Littlest JupyterHub (TLJH) on Debain 11/12 and Ubuntu 20.04 LTS/22.04 LTS – Linux Hint

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7-8 minutos

The Littlest Jupyter Hub (TLJH) is a Jupyter Hub distribution. The main goal of this project is to make it easy to set up Jupyter Hub for a small number of users (100 users at most) on a single server. TLJH allows the users to use Jupyter Lab in their own isolated environment.

In this article, we will show you how to install The Littlest Jupyter Hub (TLJH) on Debian 11, Debian 12, Ubuntu 20.04 LTS, and Ubuntu 22.04 LTS operating systems. We will also show you how to access Jupyter Hub, create new TLJH users, and manage the TLJH user sessions. We will show you how to install the new Python libraries for all TLJH users as well.

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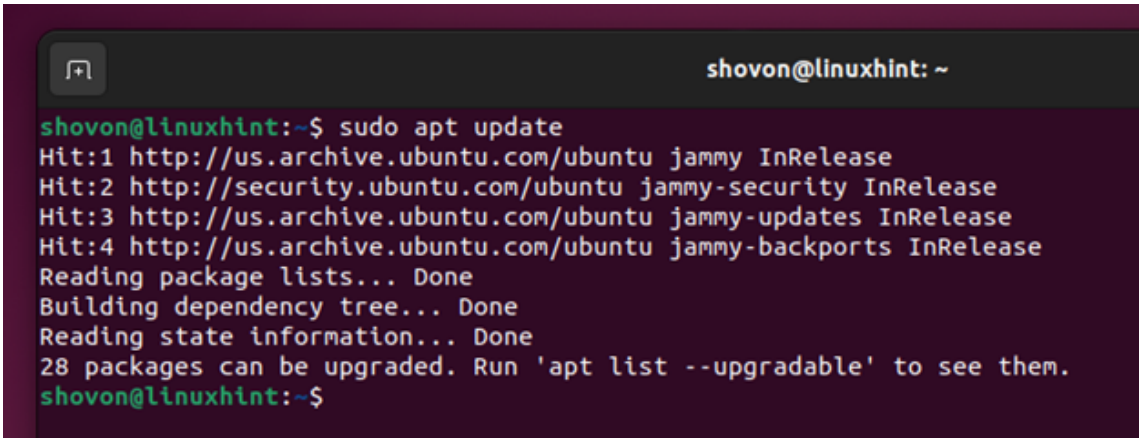
Installing the Dependency Packages for TLJH

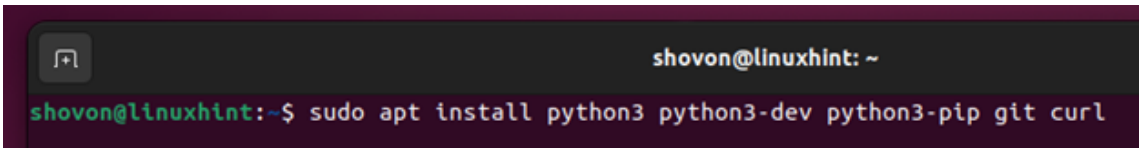
First, update the APT package database cache with the following command:

$ sudo apt update



To install the required dependency packages for Jupyter Hub, run the following command:

$ sudo apt install python3 python3-dev python3-pip git curl



To confirm the installation, press “Y” and then press **<Enter>.**

```
shovon@linuxhint: ~
  libfile-fcntllock-perl libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0
  libnsl-dev libpython3-dev libpython3.10-dev libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0
  libubsan1 linux-libc-dev lto-disabled-list make manpages-dev python3-distutils python3-setuptools
  python3-wheel python3.10-dev rpcsvc-proto zlib1g-dev
Suggested packages:
  binutils-doc debian-keyring g++-multilib g++-11-multilib gcc-11-doc gcc-multilib autoconf automake
  libtool flex bison gcc-doc gcc-11-multilib gcc-11-locales git-daemon-run | git-daemon-sysvinit
  git-doc git-email git-gui gitk gitweb git-cvs git-mediawiki git-svn apache2 | lighttpd | httpd
  glibc-doc bzr libstdc++-11-doc make-doc python-setuptools-doc
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential curl dpkg-dev fakeroot g++ g++-11
  gcc gcc-11 git git-man javascript-common libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libasan6 libbinutils libc-dev-bin libc-devtools libc6-dev libcc1-0
  libcrypt-dev libctf-nobfd0 libctf0 libdpkg-perl liberror-perl libexpat1-dev libfakeroot
  libfile-fcntllock-perl libgcc-11-dev libitm1 libjs-jquery libjs-sphinxdoc libjs-underscore liblsan0
  libnsl-dev libpython3-dev libpython3.10-dev libquadmath0 libstdc++-11-dev libtirpc-dev libtsan0
  libubsan1 linux-libc-dev lto-disabled-list make manpages-dev python3-dev python3-distutils
  python3-pip python3-setuptools python3-wheel python3.10-dev rpcsvc-proto zlib1g-dev
The following packages will be upgraded:
  libcurl4
1 upgraded, 57 newly installed, 0 to remove and 27 not upgraded.
Need to get 66.9 MB of archives.
After this operation, 242 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

The dependency packages are being downloaded. It takes a while to complete.

```
shovon@linuxhint: ~
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbinutils amd64 2.38-4ubuntu2.3 [662 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libctf-nobfd0 amd64 2.38-4ubuntu2.3 [107 kB]
Get:4 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libctf0 amd64 2.38-4ubuntu2.3 [103 kB]
Get:5 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 binutils-x86-64-linux-gnu amd64 2.38-4ubuntu2.3 [2
,327 kB]
Get:6 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 binutils amd64 2.38-4ubuntu2.3 [3,190 B]
Get:7 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-dev-bin amd64 2.35-0ubuntu3.4 [20.3 kB]
Get:8 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-libc-dev amd64 5.15.0-87.97 [1,339 kB]
Get:9 http://us.archive.ubuntu.com/ubuntu jammy/main amd64 libcrypt-dev amd64 1:4.4.27-1 [112 kB]
Get:10 http://us.archive.ubuntu.com/ubuntu jammy/main amd64 rpcsvc-proto amd64 1.4.2-0ubuntu6 [68.5 kB]
Get:11 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtirpc-dev amd64 1.3.2-2ubuntu0.1 [192 kB]
Get:12 http://us.archive.ubuntu.com/ubuntu jammy/main amd64 libnsl-dev amd64 1.3.0-2build2 [71.3 kB]
Get:13 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-dev amd64 2.35-0ubuntu3.4 [2,100 kB]
Get:14 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcc1-0 amd64 12.3.0-1ubuntu1~22.04 [48.3 kB]
Get:15 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libitm1 amd64 12.3.0-1ubuntu1~22.04 [30.2 kB]
Get:16 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasan6 amd64 11.4.0-1ubuntu1~22.04 [2,282 kB]
Get:17 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 liblsan0 amd64 12.3.0-1ubuntu1~22.04 [1,069 kB]
Get:18 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtsan0 amd64 11.4.0-1ubuntu1~22.04 [2,260 kB]
Get:19 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libubsan1 amd64 12.3.0-1ubuntu1~22.04 [976 kB]
Get:20 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libquadmath0 amd64 12.3.0-1ubuntu1~22.04 [154 kB]
Get:21 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-11-dev amd64 11.4.0-1ubuntu1~22.04 [2,517
kB]
Get:22 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11 amd64 11.4.0-1ubuntu1~22.04 [20.1 MB]
28% [22 gcc-11 948 kB/20.1 MB 5%]                                                    1,304 kB/s 37s
```

The dependency packages are being downloaded. It takes a while to complete.

```
shovon@linuxhint: ~
Unpacking libcc1-0:amd64 (12.3.0-1ubuntu1~22.04) ...
Selecting previously unselected package libitm1:amd64.
Preparing to unpack .../14-libitm1_12.3.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libitm1:amd64 (12.3.0-1ubuntu1~22.04) ...
Selecting previously unselected package libasan6:amd64.
Preparing to unpack .../15-libasan6_11.4.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libasan6:amd64 (11.4.0-1ubuntu1~22.04) ...
Selecting previously unselected package liblsan0:amd64.
Preparing to unpack .../16-liblsan0_12.3.0-1ubuntu1~22.04_amd64.deb ...
Unpacking liblsan0:amd64 (12.3.0-1ubuntu1~22.04) ...
Selecting previously unselected package libtsan0:amd64.
Preparing to unpack .../17-libtsan0_11.4.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libtsan0:amd64 (11.4.0-1ubuntu1~22.04) ...
Selecting previously unselected package libubsan1:amd64.
Preparing to unpack .../18-libubsan1_12.3.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libubsan1:amd64 (12.3.0-1ubuntu1~22.04) ...
Selecting previously unselected package libquadmath0:amd64.
Preparing to unpack .../19-libquadmath0_12.3.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libquadmath0:amd64 (12.3.0-1ubuntu1~22.04) ...
Selecting previously unselected package libgcc-11-dev:amd64.
Preparing to unpack .../20-libgcc-11-dev_11.4.0-1ubuntu1~22.04_amd64.deb ...
Unpacking libgcc-11-dev:amd64 (11.4.0-1ubuntu1~22.04) ...

Progress: [ 17%] [################.....................................................................................]
```

The dependency packages should be installed on your Ubuntu/Debian machine at this point.

```
shovon@linuxhint: ~
Setting up libjs-sphinxdoc (4.3.2-1) ...
Setting up libgcc-11-dev:amd64 (11.4.0-1ubuntu1~22.04) ...
Setting up libc6-dev:amd64 (2.35-0ubuntu3.4) ...
Setting up binutils-x86-64-linux-gnu (2.38-4ubuntu2.3) ...
Setting up binutils (2.38-4ubuntu2.3) ...
Setting up dpkg-dev (1.21.1ubuntu2.2) ...
Setting up libexpat1-dev:amd64 (2.4.7-1ubuntu0.2) ...
Setting up libstdc++-11-dev:amd64 (11.4.0-1ubuntu1~22.04) ...
Setting up zlib1g-dev:amd64 (1:1.2.11.dfsg-2ubuntu9.2) ...
Setting up gcc-11 (11.4.0-1ubuntu1~22.04) ...
Setting up g++-11 (11.4.0-1ubuntu1~22.04) ...
Setting up gcc (4:11.2.0-1ubuntu1) ...
Setting up libpython3.10-dev:amd64 (3.10.12-1~22.04.2) ...
Setting up python3.10-dev (3.10.12-1~22.04.2) ...
Setting up g++ (4:11.2.0-1ubuntu1) ...
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode
Setting up build-essential (12.9ubuntu3) ...
Setting up libpython3-dev:amd64 (3.10.6-1~22.04) ...
Setting up python3-dev (3.10.6-1~22.04) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.4) ...
shovon@linuxhint:~$
```

Installing The Littlest Jupyter Hub (TLJH)

Before you install TLJH on your Ubuntu/Debian machine, you have to find the username of the user that you want to configure as a TLJH administrator. Usually, you will configure the login user of your Ubuntu/Debian machine as a TLJH administrator unless you have something else in mind. You can configure any user as TLJH administrator depending on your requirements.

You can find the username of your login user with the following command:

$ whoami

In our case, the login user name is “shovon”.

```
shovon@linuxhint:~$ whoami
shovon
shovon@linuxhint:~$
```

To install TLJH on your Ubuntu/Debian machine, run the following command:

$ curl -L https://tljh.jupyter.org/bootstrap.py | sudo -E python3 - --admin $(whoami)

NOTE: We are configuring our login user as TLJH administrator. If you want to configure some other user as TLJH administrator, replace –admin $(whoami) with –admin in the previous command.

TLJH is being installed on your Ubuntu/Debian machine. It takes a while to download and install all the TLJH components.

```
shovon@linuxhint: ~
shovon@linuxhint:~$ curl -L https://tljh.jupyter.org/bootstrap.py | sudo -E python3 - --admin $(whoami)
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:--  0:00:01 --:--:--     0
100 18391  100 18391    0     0   8534      0  0:00:02  0:00:02 --:--:--  8534
Existing TLJH installation not detected, installing...
Setting up hub environment...
Installing Python, venv, pip, and git via apt-get...
```

At this point, TLJH should be installed on your Ubuntu/Debian machine.

```
shovon@linuxhint: ~
Setting up hub environment...
Installing Python, venv, pip, and git via apt-get...
Setting up virtual environment at /opt/tljh/hub
Upgrading pip...
Installing TLJH installer...
Running TLJH installer...
Setting up admin users
Granting passwordless sudo to JupyterHub admins...
Setting up user environment...
Downloading & setting up user environment...
Downloading conda installer https://github.com/conda-forge/miniforge/releases/download/23.1.0-1/Mambaforge-23.1.0-1-L
inux-x86_64.sh
Downloaded conda installer https://github.com/conda-forge/miniforge/releases/download/23.1.0-1/Mambaforge-23.1.0-1-Li
nux-x86_64.sh in 37.2s
/opt/tljh/user has pip==23.0.1, it will be upgraded to pip>=23.1.2
Setting up JupyterHub...
Downloading traefik 2.10.1 from https://github.com/traefik/traefik/releases/download/v2.10.1/traefik_v2.10.1_linux_am
d64.tar.gz...
Created symlink /etc/systemd/system/multi-user.target.wants/jupyterhub.service → /etc/systemd/system/jupyterhub.servi
ce.
Created symlink /etc/systemd/system/multi-user.target.wants/traefik.service → /etc/systemd/system/traefik.service.
Waiting for JupyterHub to come up (1/20 tries)
Done!
shovon@linuxhint:~$
```

Checking If TLJH Is Working

To verify that TLJH is working on your computer, check first whether the TLJH proxy service traefik is working with the following command:

$ sudo systemctl status traefik.service

As you can see, the traefik proxy service is running and it's configured to automatically start on system boot.

```
shovon@linuxhint: ~
shovon@linuxhint:~$ sudo systemctl status traefik.service
[sudo] password for shovon:
● traefik.service
     Loaded: loaded (/etc/systemd/system/traefik.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2023-10-19 10:16:01 MDT; 1h 26min ago
   Main PID: 5500 (traefik)
      Tasks: 9 (limit: 4556)
     Memory: 20.7M
        CPU: 551ms
     CGroup: /system.slice/traefik.service
             └─5500 /opt/tljh/hub/bin/traefik -c /opt/tljh/state/traefik.toml

Oct 19 10:16:01 linuxhint systemd[1]: Started traefik.service.
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Configuration loaded from >
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Traefik version 2.10.1 bui>
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="\nStats collection is disa>
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Starting provider aggregat>
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Starting provider *file.Pr>
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Starting provider *traefik>
Oct 19 10:16:01 linuxhint traefik[5500]: time="2023-10-19T10:16:01-06:00" level=info msg="Starting provider *acme.Ch>
Oct 19 10:26:03 linuxhint traefik[5500]: time="2023-10-19T10:26:03-06:00" level=warning msg="A new release has been >
lines 1-19/19 (END)
```

Also, check whether the JupyterHub service is working with the following command:

$ sudo systemctl status jupyterhub.service

As you can see, the JupyterHub service is also running and it's configured to automatically start on system boot.

```
shovon@linuxhint: ~
shovon@linuxhint:~$ sudo systemctl status jupyterhub.service
● jupyterhub.service
     Loaded: loaded (/etc/systemd/system/jupyterhub.service; enabled; vendor preset: enabled)
     Active: active (running) since Thu 2023-10-19 10:16:01 MDT; 1h 26min ago
   Main PID: 5501 (python3)
      Tasks: 3 (limit: 4556)
     Memory: 76.3M
        CPU: 1.536s
     CGroup: /system.slice/jupyterhub.service
             ├─5501 /opt/tljh/hub/bin/python3 -m jupyterhub.app -f /opt/tljh/hub/lib/python3.10/site-packages/tljh/j>
             └─5560 /opt/tljh/hub/bin/python3 -m jupyterhub_idle_culler --timeout=600 --cull-every=60 --concurrency=>

Oct 19 11:38:02 linuxhint python3[5501]: [I 2023-10-19 11:38:02.854 JupyterHub log:191] 200 GET /hub/api/ (cull-idle>
Oct 19 11:38:02 linuxhint python3[5501]: [I 2023-10-19 11:38:02.856 JupyterHub log:191] 200 GET /hub/api/users?state>
Oct 19 11:39:02 linuxhint python3[5501]: [I 2023-10-19 11:39:02.836 JupyterHub log:191] 200 GET /hub/api/ (cull-idle>
Oct 19 11:39:02 linuxhint python3[5501]: [I 2023-10-19 11:39:02.839 JupyterHub log:191] 200 GET /hub/api/users?state>
Oct 19 11:40:02 linuxhint python3[5501]: [I 2023-10-19 11:40:02.859 JupyterHub log:191] 200 GET /hub/api/ (cull-idle>
Oct 19 11:40:02 linuxhint python3[5501]: [I 2023-10-19 11:40:02.862 JupyterHub log:191] 200 GET /hub/api/users?state>
Oct 19 11:41:02 linuxhint python3[5501]: [I 2023-10-19 11:41:02.811 JupyterHub log:191] 200 GET /hub/api/ (cull-idle>
Oct 19 11:41:02 linuxhint python3[5501]: [I 2023-10-19 11:41:02.814 JupyterHub log:191] 200 GET /hub/api/users?state>
Oct 19 11:42:02 linuxhint python3[5501]: [I 2023-10-19 11:42:02.810 JupyterHub log:191] 200 GET /hub/api/ (cull-idle>
Oct 19 11:42:02 linuxhint python3[5501]: [I 2023-10-19 11:42:02.813 JupyterHub log:191] 200 GET /hub/api/users?state>
lines 1-21/21 (END)
```

As the traefik proxy and JupyterHub "systemd" services are running correctly, The Littelest Jupyter Hub (TLJH) is working just fine.

Accessing TLJH

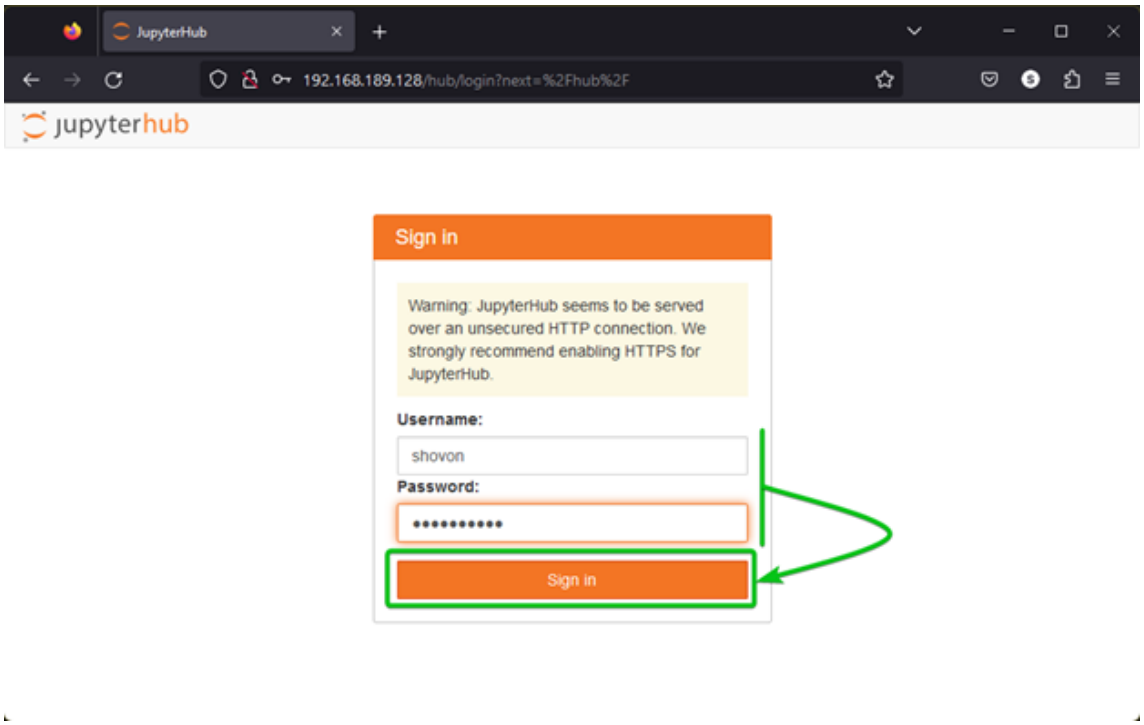
To access TLJH from a web browser, you need to know the IP address (or DNS name if configured) of your Ubuntu/Debian machine. In our case, the IP address is 192.168.189.128. It might be different for you. So, make sure to replace it with yours from now on.

$ ip a

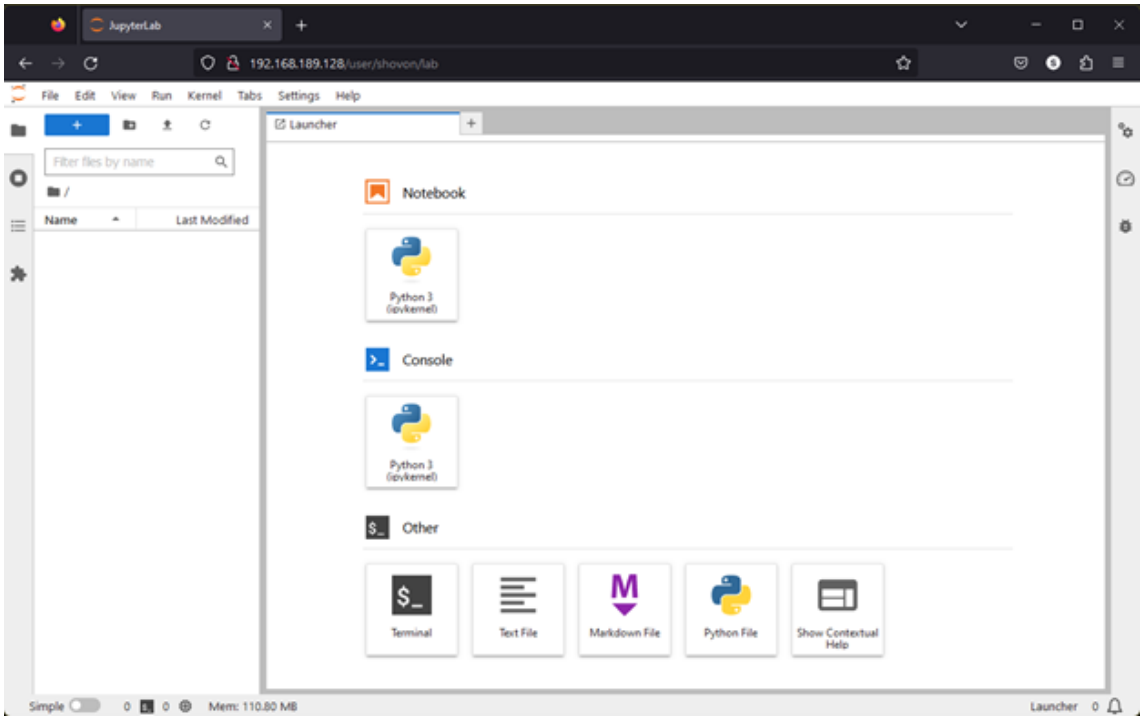
```
shovon@linuxhint: ~
shovon@linuxhint:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:4c:d7:de brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.189.128/24 brd 192.168.189.255 scope global dynamic noprefixroute ens33
       valid_lft 1097sec preferred_lft 1097sec
    inet6 fe80::a8be:d5f8:78bc:3393/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
shovon@linuxhint:~$
```

Open a web browser app and visit http://192.168.189.128 and you will see the JupyterHub login page.

Type in your admin username, your desired login password (that you want to set for the admin user of JupyterHub), and click on "Sign in".

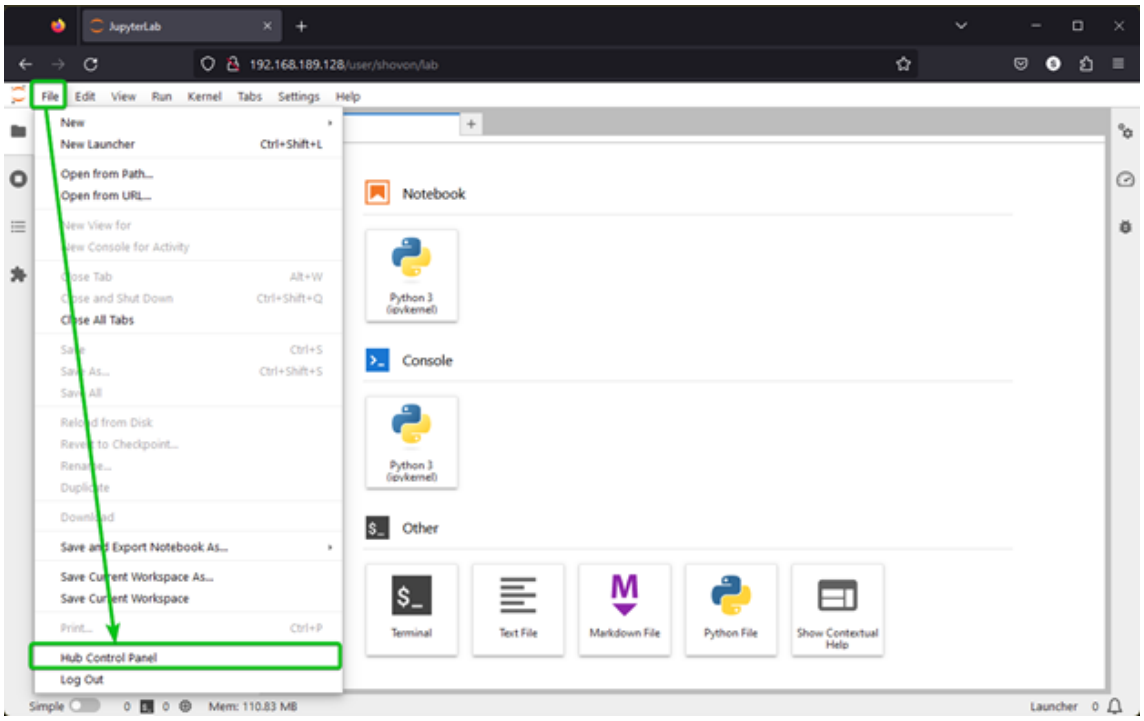


You should be logged in to JupyterHub.

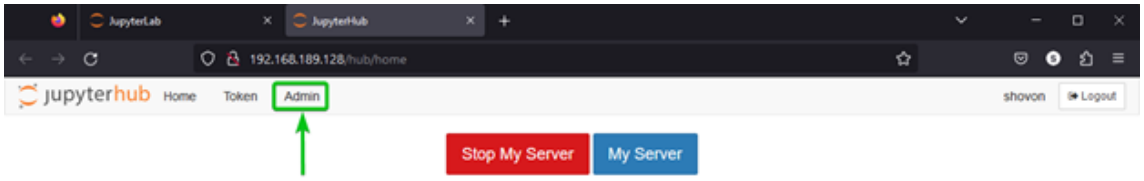


Creating TLJH Users and Managing the TLJH User Sessions

To create new TLJH users, log in to JupyterHub as the admin user and click on **File > Hub Control Panel.**

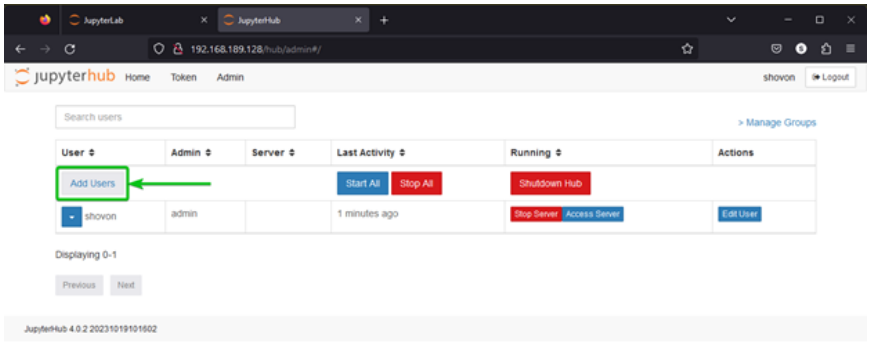


Click on “Admin”.



All the TLJH users and user sessions/hubs should be listed.

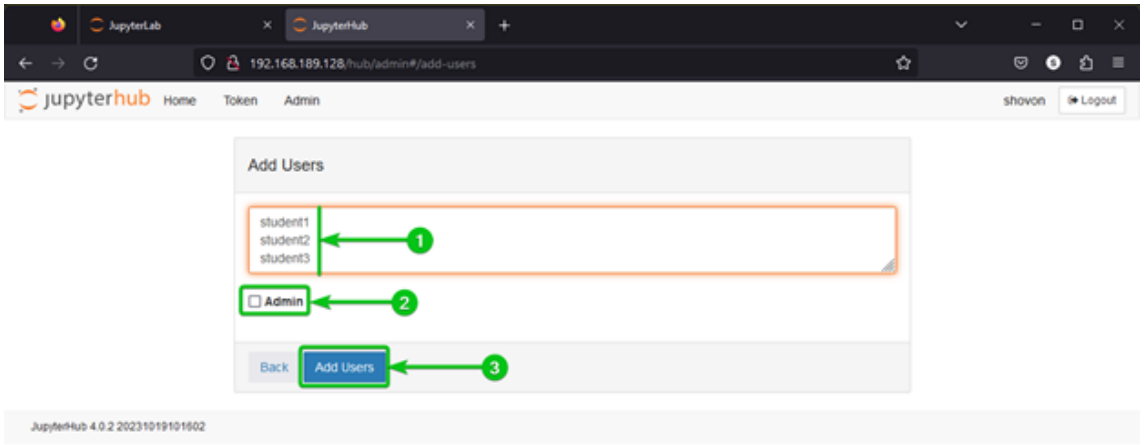
Click on “Add Users”.



Type in the usernames that you want to add to TLJH[1]. You can add a single user or multiple users to TLJH at once. To add multiple users to TLJH, type in each username in a separate line.

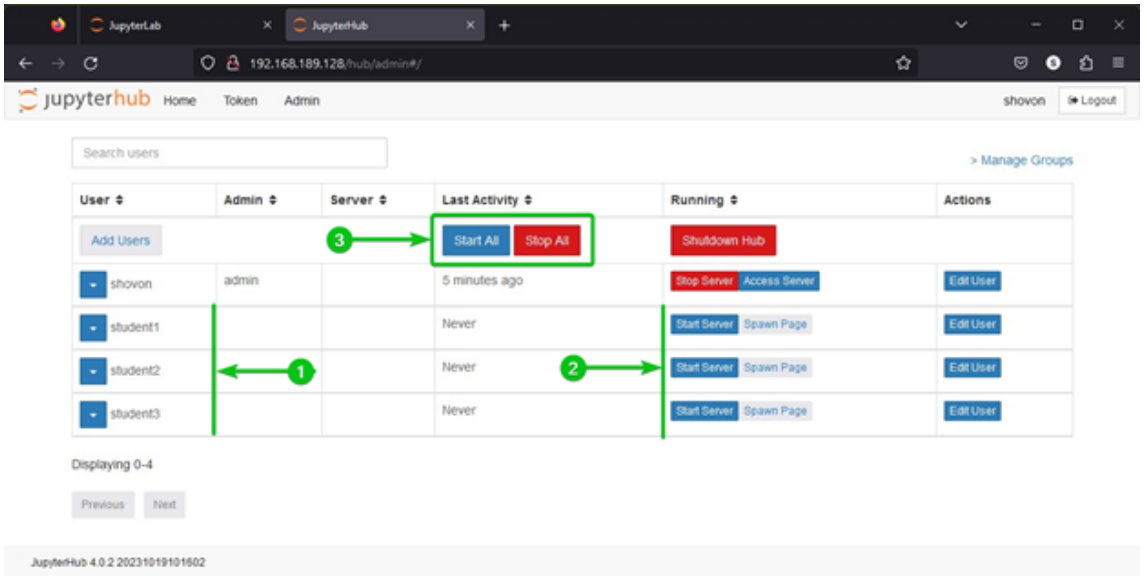
If you want the newly created users to have administrative access to TLJH, tick on “Admin”[2].

Once you’re done, click on “Add Users”[3].



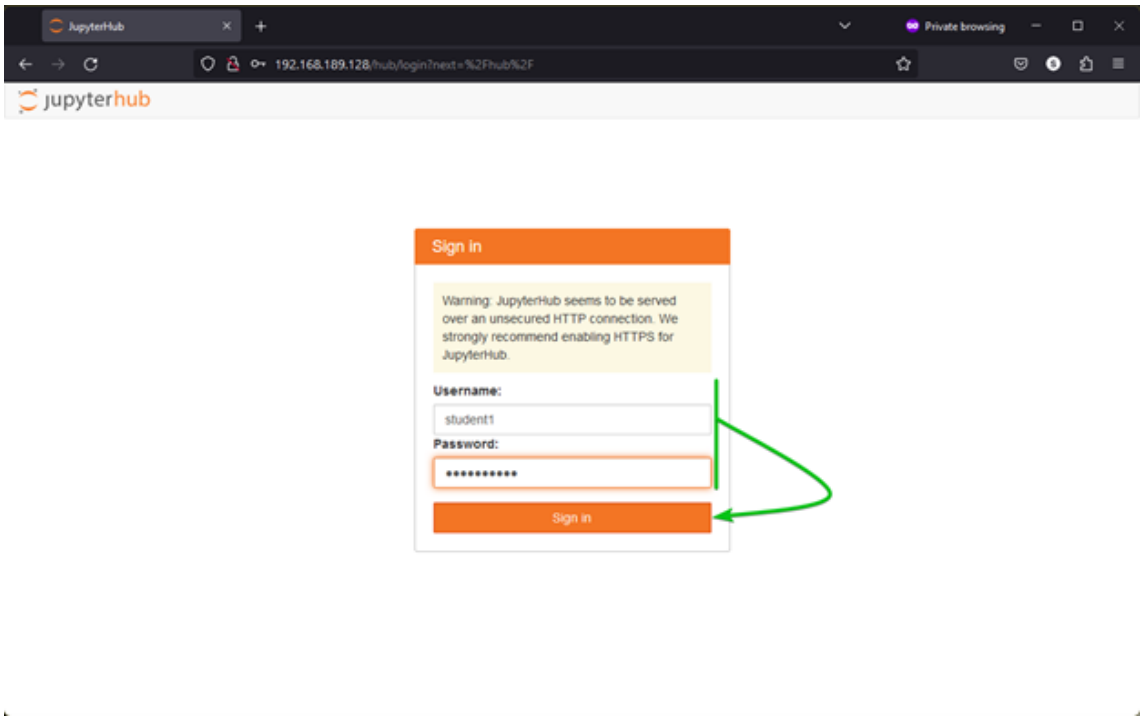
The new TLJH users should be created[1].

From the “Admin” page, you can manage (Start/Stop) the TLJH user sessions/hubs for each user[2] and all users[3].

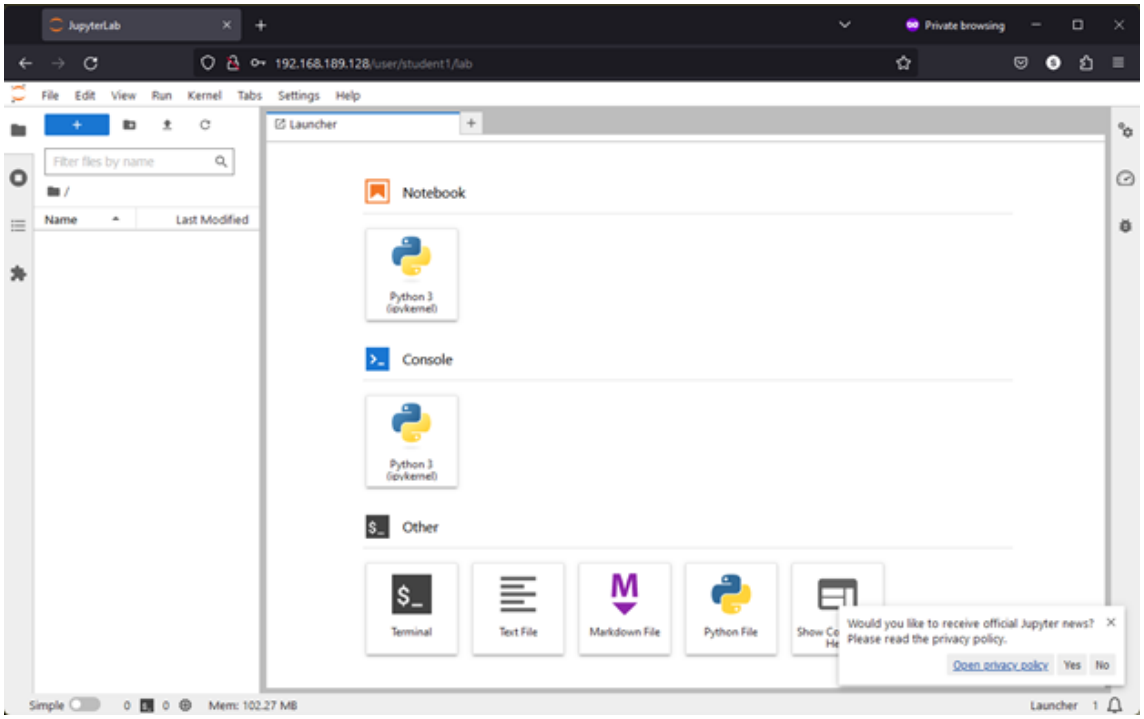


To log in as one of the TLJH users, type in the username and the password that you want to set for the new user and click on “Sign in”.

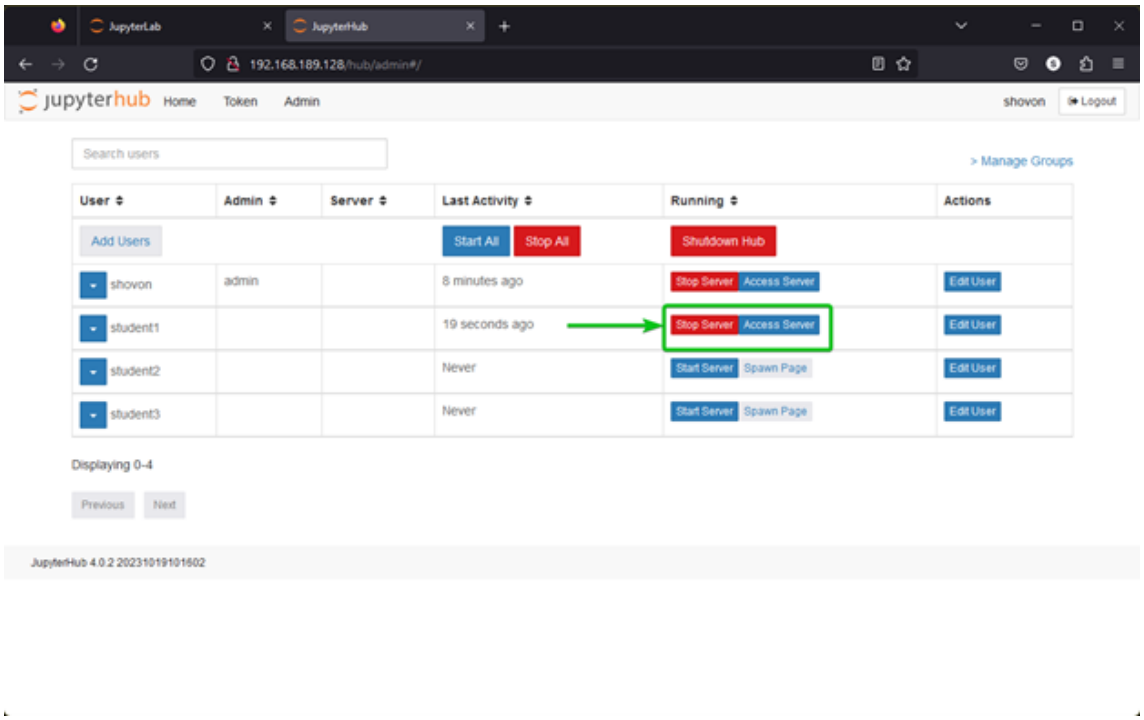
NOTE: The password that you use to log in for the first time will be set as the login password for the new user.



You should be logged in to JupyterHub as the new user.



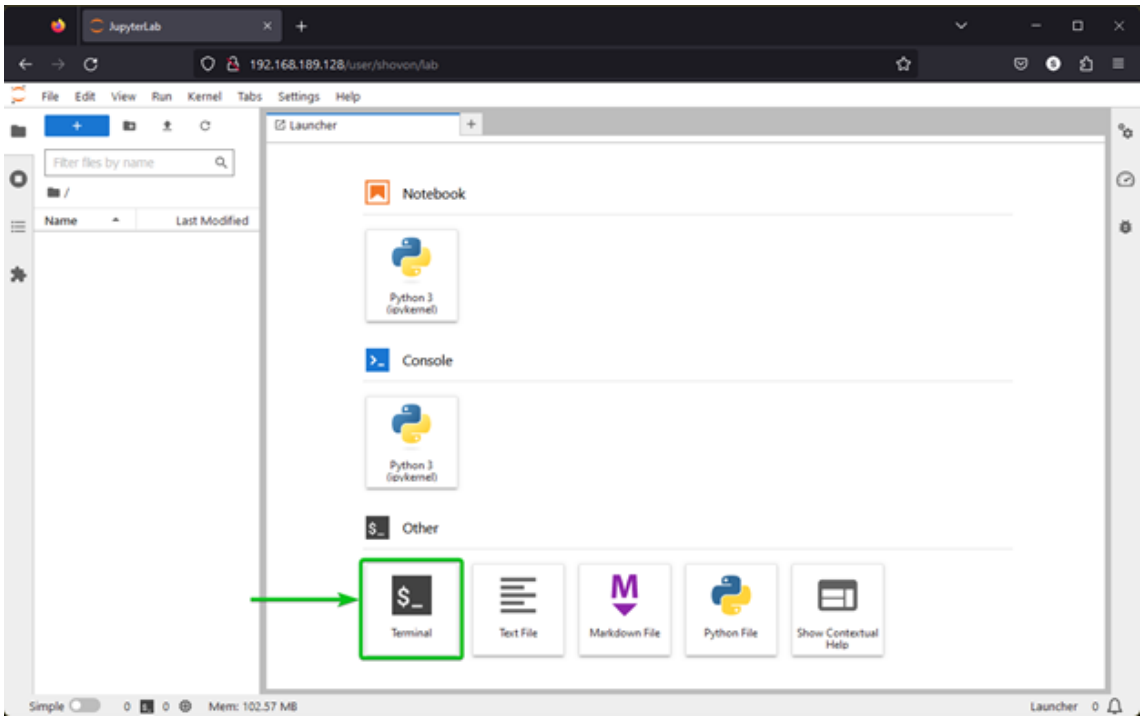
The TLJH admin user can see the users who are logged in to TLJH and manage the user sessions/hub.



Installing the Python Libraries for All Jupyter Hub Users

Each TLJH user can install any Python libraries with Python PIP from their Jupyter Hub session. But to save the disk space, you can install the most commonly used Python libraries system-wide so that all Jupyter Hub users can access them without needing to reinstall them.

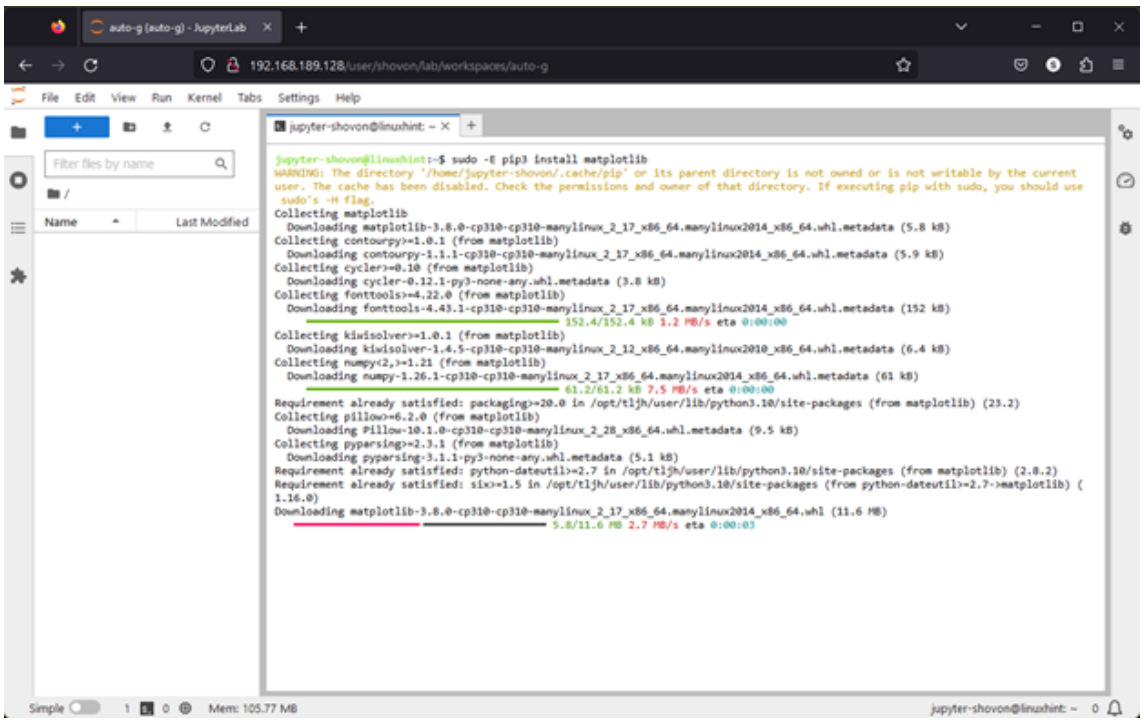
First, log in to TLJH as the admin user and click on “Terminal” from the “Launcher” tab.



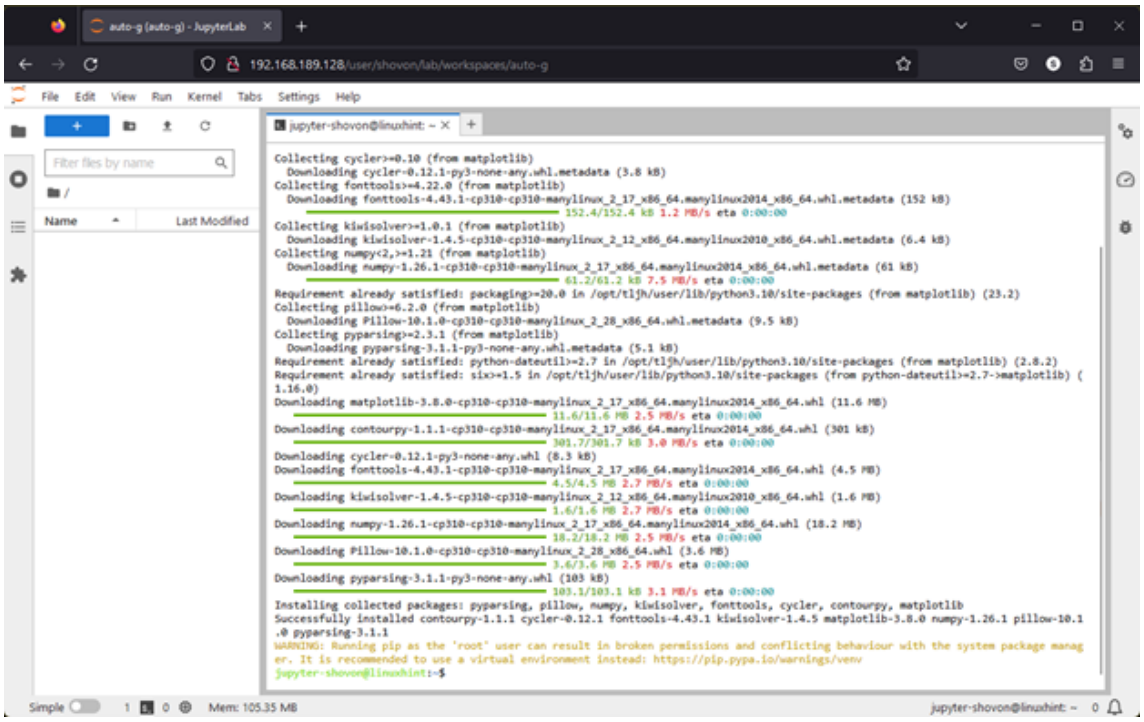
To install the matplotlib Python library (let’s say) for all TLJH users, run the following command:

$ sudo -E pip3 install matplotlib

Matplotlib is being installed. It takes a while to complete.



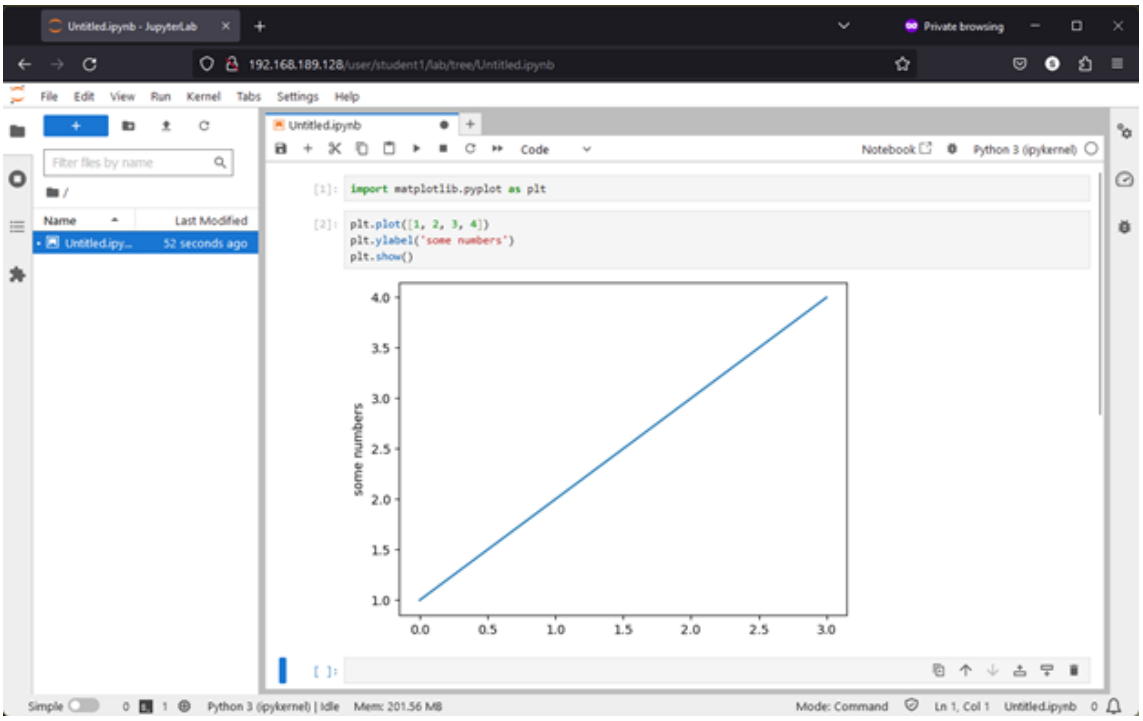
At this point, the matplotlib Python library should be installed for all TLJH users.



To check whether other TLJH users can use the matplotlib Python library, login as one of the TLJH users, create a new Jupyter Notebook, and run the following lines of codes (that we copied from the matplotlib documentation):

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4])
plt.ylabel('some numbers')
plt.show()
```

If the TLJH user can access matplotlib, the code should generate a graph as shown in the following screenshot:



Conclusion

In this article, we showed you how to install The Littlest Jupyter Hub (TLJH) on Debian 11, Debian 12, Ubuntu 20.04 LTS, and Ubuntu 22.04 LTS operating systems. We also showed you how to access TLJH from a web browser, create new TLJH users, and manage the TLJH user sessions. We showed you how to install the new Python libraries for all TLJH users as well.

References:

- Installing on your own server — The Littlest JupyterHub documentation